PARABOLIC DYNAMICS AND ANISOTROPIC BANACH SPACES

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Abstract. We explain the relation between the distributions appearing in the study of ergodic averages of parabolic flows (e.g. in the work of Forni) and the ones appearing in the study of the statistical properties of hyperbolic dynamical systems (i.e. the eigendistributions of the transfer operator). In order to avoid, as much as possible, technical issues that would cloud the basic idea, we limit ourselves to the simplest possible example. Nevertheless, the conceptual connection that we illustrate is expected to hold in considerable generality and should allow to substantially improve the current technology.

1. Introduction

In the last decade, *distributions* have became increasingly relevant both in parabolic and hyperbolic dynamics. On the *parabolic dynamics* side consider, for example, the work of Forni [15, 12, 13, 14] on ergodic averages and cohomological equations for horocycle flows; on the *hyperbolic dynamics* side it suffices to mention the study of the transfer operator through anisotropic spaces, started with [8].[1]

Since a typical approach to the study of *parabolic dynamics* is the use of renormalization techniques,[2] where the renormalizing dynamics is typically a *hyperbolic dynamics*, several people have been wondering on a possible relation between such two classes of distributions.

In this paper we show that the distributional obstructions in Forni's work and the distributional eigenvectors of certain transfer operators are tightly related, to the point that, informally, we could say that they are exactly the same.

In order to present our argument in the simplest possible manner, instead of trying to develop it for the horocycle flow versus the geodesic flow (which would require a more technical framework), we consider a very simple example that preserves the main ingredients of the horocycle-geodesic flow setting and, at the same time, shows the flexibility of the argument itself, which has the potential of being greatly generalized.

As parabolic dynamics, we consider a flow ϕ_t, over $\mathbb{T}^2 = {}^{\mathbb{R}^2}/_{\mathbb{Z}^2}$, generated by a vector field $V \in \mathcal{C}^{1+\alpha}(\mathbb{T}^2, \mathbb{R}^2)$, $\alpha \in \mathbb{R}_{\geq}$, such that, $\forall x \in \mathbb{T}^2$, $V(x) \neq 0$. As the hyperbolic dynamics we consider a transitive Anosov map $F \in \mathcal{C}^r(\mathbb{T}^2, \mathbb{T}^2)$, $r > 1+\alpha$.

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[1] But see, e.g., [27, 28, 26, 25] for earlier related results.

[2] Typical examples are circle rotations [31, 32], interval exchange maps via Teichmuller theory [33, 1], horocycle flow [15, 12].

We call $E^s(x)$ and $E^u(x)$ the corresponding stable and unstable subspaces at each $x \in \mathbb{T}^2$. As we want the latter to act as a renormalizing dynamics for the former, we require, $\forall x \in \mathbb{T}^2$,

$$V(x) \in E^s(x). \tag{1.1}$$

Remark 1.1. *In general the distribution E^s will be only $\mathcal{C}^{1+\alpha}$ with $\alpha \in (0,1)$, [24]. However, is it possible to have situations in which $\alpha > 1$ and yet F is not $\mathcal{C}^1$ conjugated to a toral automorphism* [24, Exercise 19.1.5]. *Of course, in the latter case the unstable foliation will be irregular* [18, Corollary 3.3]. *One could argue that it is always possible to make a $\mathcal{C}^{1+\alpha}$ change of variables to conjugate V with a constant vector field and then use a toral automorphism as a renormalizing dynamics. In such a case however one would be naturally restricted to considering $\mathcal{C}^{1+\alpha}$ rather than $\mathcal{C}^r$ regularity, which, as we will see shortly, is essential for the questions we are interested in. See also Remark 1.4 and the end of Remark 1.7 for further considerations.*

Equation (1.1) implies that the trajectories $\gamma = \{\phi_t(x)\}_{t\in(a,b)}$ are pieces of the stable manifolds for the map F. Moreover, $D_xF^nV(x) = \nu_n(x)V(F^n(x))$, where $|\nu_n| < \lambda^{-n}$ for some $\lambda > 1$. Without loss of generality we assume that F preserves the orientation of the invariant splittings, i.e. $\nu_n > 0$ (if note, use F^2).

Given the hypothesis (1.1) it is natural to ask, at least, that, for each $x \in \mathbb{R}^2$,

$$\phi_{(\cdot)}(x) \in \mathcal{C}^r. \tag{1.2}$$

Remark 1.2. *Note that if $\|V(x)\| = 1$ for all $x \in \mathbb{T}^2$, then hypothesis (1.2), is automatically satisfied due to hypothesis (1.1). Indeed, being $F \in \mathcal{C}^r$, so are the stable leaves* [24]. *We will however need a stronger condition, see Definition 2.3 (ii) and Remark 2.4.*

Remark 1.3. *The reader may complain that the parabolic nature of the dynamics ϕ_t it is not very apparent. Indeed, a little argument is required to show that $\|D_x\phi_t\|$ can grow at most polynomially in t (see Lemma 4.2).*

Let us better detail in which sense F renormalises ϕ_t. Given $x \in \mathbb{T}^2$, for each $n \in \mathbb{N}$ and $t \in \mathbb{R}_\geq$, we define

$$\tau_n(x,t) = \int_0^t ds\, \nu_n(\phi_s(x)). \tag{1.3}$$

By construction, for each $x \in \mathbb{T}^2$ and $n \in \mathbb{N}$, $\tau_n(x,t)$ is a strictly increasing function of t, and hence globally invertible. We will use the, slightly misleading, notation $\tau_n^{-1}(x,\cdot)$ for the inverse. Next, letting $\gamma(x,t) = \phi_t(x)$ we define

$$\gamma_n(x,\tau_n(x,t)) = F^n(\gamma(x,t)) = F^n(\phi_t(x)) \tag{1.4}$$

By direct computation we have

$$\gamma_n'(x,\tau_n(x,t))\nu_n(\phi_t(x)) = D_{\phi_t(x)}F^nV(\phi_t(x)) = \nu_n(\phi_t(x))V(\gamma_n(x,\tau_n(x,t))).$$

Accordingly,

$$\begin{cases} \frac{d}{ds}\gamma_n(x,s) = V(\gamma_n(x,s)) \\ \gamma_n(x,0) = F^n(x) \end{cases} \tag{1.5}$$

Thus, by the uniqueness of the solution of the above ODE, it follows $\gamma_n(x,s) = \phi_s(F^n(x)) = \gamma(F^n(x), s)$. In other words, the image under F^n of a piece of trajectory, is the reparametrization of a (much shorter) piece of trajectory. More precisely, we can write

$$F^n(\phi_t(x)) = \phi_{\tau_n(x,t)}(F^n(x)). \tag{1.6}$$

Next, note that, by hypotheses, the flow ϕ_t does not have fixed points nor periodic orbits.[3] Hence there exists a global section uniformly transversal to the flow[4] and the associated Poincaré map is a $\mathcal{C}^{1+\alpha}$ map of the circle with irrational rotation number. It follows by Furstenberg [17] that the Poincaré map is uniquely ergodic, hence the same holds for the flow. Let μ be the unique invariant measure.

By unique ergodicity, given $g \in \mathcal{C}^0(\mathbb{T}^2, \mathbb{R})$, $\frac{1}{t}\int_0^t ds\, g \circ \phi_s(x)$ converges uniformly to $\mu(g)$. We have thus naturally arrived at our

First question: How fast is the convergence to the ergodic average?

The question is equivalent to investigating the precise growth of

$$H_{x,t}(g) := \int_0^t ds\, g \circ \phi_s(x). \tag{1.7}$$

Remark 1.4. *Note that Denjoy Theorem [24] implies that the Poincaré map is topologically conjugated to a rigid circle rotation. However in general, even for Diophantine rotations number, the Poicaré map may not be $\mathcal{C}^1$ conjugated due to the (possible) low regularity of the map that does not allow to apply KAM type strategies. Accordingly, the study of the rate of convergence to the ergodic averages cannot, in general, be easily reduced to the rigid case in which ϕ_t is a translation and F a toral automorphism.*

In analogy with the work of Forni one expects that there exist a finite number of functionals $\{O_i\}_{i=1,\dots,N_1} \subset \mathcal{C}^r(\mathbb{T}^2,\mathbb{R})'$, and a set $\{\alpha_i\}_{i=1,\dots,N_1}$ of decreasing numbers $\alpha_i \in [0,1]$ such that if $O_j(g) = 0$ for all $j < i$ and $O_i(g) \neq 0$, then $H_{x,t}(g) = \mathcal{O}(t^{\alpha_i})$. As we have seen, $O_1(g) = \mu(g)$ with $\alpha_1 = 1$.[5] On the other hand, unless $g \equiv 0$, the integral cannot converge to zero, hence we can have, at most, that $\alpha_{N_1} = 0$.

A remarkable discovery of Forni is that the functionals above cannot be expected, in general, to be measures: they are *distributions*.

Next, suppose that $O_i(g) = 0$ for all $i \leq N_1$, $\alpha_{N_1} = 0$, then $H_{x,t}(g)$ remains bounded. It is then convenient to introduce, for each $(x,T) \in \mathbb{T}^2 \times \mathbb{R}_{\geq}$ the new functionals $\overline{H}_{x,T} : \mathcal{C}^r \to \mathbb{R}$ defined by

$$\overline{H}_{x,T}(g) := -\int_0^T dt\, \chi \circ \tau_{n_T}(x,t) H_{x,t}(g), \tag{1.8}$$

[3] A periodic orbit would imply that an invariant leaf of F is compact, which is impossible.

[4] See [30] for the original work, or [19] for a brief history of the problem and references.

[5] We will see that μ is also the maximal left eigenvector of some transfer operator associated to F. In fact, the "product" of the left and right maximal eigenvectors for such an operator yields the measure of maximal entropy for F, see [22, Section 7.0.1] for a detailed discussion and [22, Lemma 6.1] for the proof that the left maximal eigenvector is indeed a measure.

where $n_T + 1 = \inf\{n \in \mathbb{N} \, : \, \tau_n(x, T) \le 1\}$ and $\chi \in \mathcal{C}^r(\mathbb{R}_\ge, [0, 1])$ is a fixed function such that $\chi(0) = 1$, $\frac{d^k}{dt^k}\chi(0) = 0$ for all $0 \le k \le r$, and $\chi(s) = 0$ for all $s \ge 1$.[6] Observe that

$$
\begin{aligned}
\langle V(x), \nabla \overline{H}_{x,T}(g)\rangle &= -\int_0^T dt\, \chi \circ \tau_{n_T}(x,t) \langle D_x\phi_t V(x), (\nabla g)\circ \phi_t(x)\rangle \\
&= -\int_0^T dt\, \chi \circ \tau_{n_T}(x,t) \langle V(\phi_t(x)), (\nabla g)\circ \phi_t(x)\rangle \\
&= -\int_0^T dt\, \chi \circ \tau_{n_T}(x,t) \left(\frac{d}{dt} g\circ\phi_t(x)\right) \\
&= g(x) + \int_0^T dt\, \chi' \circ \tau_{n_T}(x,t)\nu_{n_T}(\phi_t(x)) g\circ\phi_t(x),
\end{aligned}
\tag{1.9}
$$

where we have used the fact that

$$
D_x\phi_s(V(x)) = D_x\phi_s \left.\frac{d}{d\tau}\phi_\tau(x)\right|_{\tau=0} = \left.\frac{d}{d\tau}\phi_{s+\tau}(x)\right|_{\tau=0} = V(\phi_s(x)). \tag{1.10}
$$

We will see (in Lemma 4.1) that if $H_{x,t}(g)$ is uniformly bounded, so is $\overline{H}_{x,T}(g)$. Then, the sequence $\{\overline{H}_{x,T}(g)\}_{T\in\mathbb{R}_\ge}$ is weakly compact in $(L^1)' = L^\infty$. Assume for a moment that rightmost term of the last line of (1.9) goes uniformly to zero, then we can rewrite (1.9) as

$$
\overline{H}_{\phi_t(x),T}(g) - \overline{H}_{x,T}(g) = \int_0^t g\circ\phi_s(x) ds + o(1).
$$

Let $h \in L^\infty$ be an accumulation point of $\{\overline{H}_{x,T}(g)\}_{T\in\mathbb{R}_\ge}$, then, for all $\varphi \in L^1(\mathbb{T}^2 \times \mathbb{R})$ with compact support,

$$
\int_{\mathbb{T}^2\times\mathbb{R}} dxdt\, \varphi(x,t)\left(h\circ\phi_t(x) - h(x) - \int_0^t g\circ\phi_s(x)ds\right) = 0. \tag{1.11}
$$

Thus $h\circ\phi_t - h = \int_0^t g\circ\phi_s ds$ for almost all x and t. It then follows, eventually by changing h on a zero measure set, that h is almost surely derivable in the flow direction. Hence,

$$
g(x) = \langle V(x), \nabla h(x)\rangle. \tag{1.12}
$$

That is, g is a measurable coboundary.

We are left with the discussion of the last term in (1.9). Note that if we would have chosen $\chi\circ\tau_{n_T} = \frac{s-T}{T}$, then such a last term would simply read

$$
\frac{1}{T}\int_0^T dt\, g\circ\phi_t(x),
$$

which would converge uniformly to zero since $\mu(g) = 0$. Unfortunately, such a choice would create catastrophic problems later on as, in the decomposition discussed in Lemma 3.1, we would end up with an integral of the type $\int_{t-\delta}^t \varphi(s) g\circ\phi_s(x)ds$ whose derivative with respect to x would yield a polynomially growing factor owing to $D_x\phi_s$. Instead our argument needs for such left overs to stay bounded. In order to avoid such a future problem we have introduced the function χ which has the property to be smooth at the left boundary of the integral. The price that we pay is

[6] The condition for $s \ge 1$ is necessary to avoid, in the following constructions, a term that cannot be controlled by our technique.

that the last term in equation (1.9) is not automatically vanishing. Yet, in Lemma 4.1 we will see that it vanishes provided g has zero average with respect to the SRB measure of F^{-1}. Hence, we count the SRB measure μ_{SRB} as the N_1+1 obstruction in our list. In other words, if g is in the kernel of $\{O_i\}_{i=1}^{N_1+1}$, then not only g is a measurable coboundary, but it can be approximated by a sequence with convenient extra properties.[7]

The above fact is of debatable interest; on the contrary, the existence of more regular solutions of (1.12) is of considerable interest and it plays a role in establishing many relevant properties (see [24, Sections 2.9, 19.2]). We arrive then at our

Second question: How regular really are the solutions of the cohomological equation (1.12)?

Following Forni again, we expect that there exist finitely many distributional obstructions $\{O_i\}_{i=N_1+2}^{N_2} \subset \mathcal{C}^r(\mathbb{T}^2,\mathbb{R})'$ and a set of increasing numbers $\{r_i\}_{i=N_1+2}^{N_2}$, $r_i \in (0, 1+\alpha)$ such that, if $O_j(g) = 0$ for all $j < i$ and $O_i(g) \neq 0$, then $h \in \mathcal{C}^{r_i}$.

Remark 1.5. *Note that in the present context, as the flow is only $\mathcal{C}^{1+\alpha}$, it is not clear if it makes any sense to look at coboundaries better than $\mathcal{C}^{1+\alpha}$. This reflects the fact that if one looks at the horocycle flows on manifolds of non constant negative curvature, then the associate vector field is, in general, not very regular. On the other hand rigidity makes not so interesting our simple example when both foliations are better than $\mathcal{C}^2$,* [18, Corollary 3.3]. *We will therefore limit ourself to finding distributions that are obstruction to Lipschitz coboundaries, i.e. if $O_i(g) = 0$ for all $i \leq N_2$, then h is Lipschitz. We believe this to be more than enough to illustrate the scope of the method.*

The goal of this paper is to prove the above facts by studying transfer operators associated to F, acting on appropriate spaces of distributions. In fact, we will show that the above mentioned obstructions $\{O_i\}$ can be derived from the eigenvectors of an appropriate transfer operator associated to F. As announced, this discloses the connection between the appearance of distributions in such two seemingly different fields of dynamical systems.

Remark 1.6. *As already mentioned, in the above setting ϕ_t plays the role of the horocycle flow, while F the one of the geodesic flow. It is important to notice that most of the results obtained for the horocycle flows (and Forni's results in particular) rely on representation theory, thus requiring constant curvature of the space. In our context, this would correspond to the assumption that F is a toral automorphism and ϕ_t a rigid translation. One could then do all the needed computations via Fourier series (see the Remark below). It is then clear that extending our approach to geodesic flows (which should be quite possible using the results on flows by* [20, 11, 29]*) would allow to treat cases of variable negative curvature, and, more generally, cases where the tools of representation theory are not available or effective, whereby greatly extending the scope of the theory.*

[7] We are not sure if such properties are really necessary or are an artefact of our method of proof. In any case we cannot get any inspiration from the existing work on the geodesic versus horocycle flow since, in the case of constant curvature, the invariant measure μ of the horocycle flow and the SRB measure of the geodesic flow both coincide with the Liouville measure.

Remark 1.7. *Let $A \in SL(2,\mathbb{N})$; let $F_A : \mathbb{T}^2 \to \mathbb{T}^2$ be the Anosov map defined by $F_A(x) \doteq Ax \mod 1$. Since $\det(A) = 1$ the map is invertible, and has eigenvalues $\lambda_A, \lambda_A^{-1} \in \mathbb{R}$, with $\lambda_A > 1$. Let $V_A = (1, \omega)$ be the eigenvector associated to the eigenvalue λ_A^{-1}. Note that ω is a quadratic irrational. Let $\phi_t(x) = x + tV_A \mod 1$. In this case by applying Fourier transform to equation (1.12) we obtain, for $k \in \mathbb{Z}^2$, calling $\hat{f}_k$ the Fourier coefficients of f,*

$$i2\pi\langle V, k\rangle \sum_{k\in\mathbb{Z}^2} \hat{h}_k e^{2\pi i\langle k,x\rangle} = \sum_{k\in\mathbb{Z}^2} \hat{g}_k e^{2\pi i\langle k,x\rangle}.$$

Note that we have the trivial obstruction $\hat{g}_k = 0$. If this is satisfied, note that $\langle V, k\rangle = k_1 + \omega k_2 \neq 0$ for all $(k_1, k_2) \in \mathbb{Z}^2 \setminus \{0\}$, since ω is irrational. Thus, we can write

$$\hat{h}(k) = -i\frac{\hat{g}(k)}{2\pi\langle V, k\rangle}.$$

Since ω is a quadratic irrational, it is well known (e.g. by using standard results on continuous fractions) that $|\langle V, k\rangle| \geq C_\# \|k\|^{-1}$. Hence, if $g \in W^{r,2}$ (the Sobolev space with the first r derivatives in L^2), then $h \in W^{r-1,2}$. In particular, if $g \in \mathcal{C}^\infty$, then $h \in \mathcal{C}^\infty$.

That is, in such simple example all the aforementioned distributions do not exist and the only obstruction is the trivial one: the one given by the invariant measure.[8]

The plan of the paper is as follow: in section 2 we state our precise assumptions, outline our reasoning and state precisely our results, assuming lemmata and constructions which are explained later on. Section 3 is devoted to our first question and proves our Theorem 2.8 concerning the distributions arising from the study of the ergodic averages. Section 4 deals with our second question and proves our Theorem 2.9 dealing with the distributions arising from the study of the regularity of the cohomological equation. In sections A and B we recall the definition of the various functional spaces used (adapted to the present simple case).

Notation. At times we will use $C_\#$ to designate a generic constant, depending only on F and ϕ_1, whose actual value can change from one occurrence to the next.

2. Definitions and main Results

Let $\alpha, r \in \mathbb{R}_\geq$ with $r > 1 + \alpha$.

We start by recalling the definition of $\mathcal{C}^r$ Anosov map of the torus.

Definition 2.1. *Let $F \in \mathcal{C}^r(\mathbb{T}^2, \mathbb{T}^2)$ where $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$. The map is called Anosov if there exists two continuous closed nontrivial transversal cone fields $C^{u,s} : \mathbb{T}^2 \to \mathbb{R}^2$ which are DF-invariant. That is, for each $x \in \mathbb{T}^2$,*

$$\begin{aligned} &D_xFC^u(x) \subset C^u(F(x)) \\ &D_xF^{-1}C^s(x) \subset C^s(F^{-1}(x)). \end{aligned} \tag{2.1}$$

In addition, there exists $C > 0$ and $\lambda > 1$ such that, for all $n \in \mathbb{N}$,

$$\begin{aligned} &\|D_xF^{-n}v\| &> C\lambda^n\|v\| \qquad &\text{if } v \in C^s(x); \\ &\|D_xF^n v\| &> C\lambda^n\|v\| \qquad &\text{if } v \in C^u(x). \end{aligned} \tag{2.2}$$

---

[8] Remark that this implies that if $\alpha > 1$, then (by the above computation and Remark 1.1) we expect $N_1 = N_2 = 1$. This seems to imply some interesting information on the spectra of the operators we are going to consider when one of the two foliations of F is regular. An issue that we have not investigated but could be of some relevance.

It is well known that the above implies the following, seemingly stronger but in fact equivalent [24], definition

Definition 2.2. *Let $F \in \mathcal{C}^r(\mathbb{T}^2, \mathbb{T}^2)$. The map is called* Anosov *if there exists a DF-invariant $\mathcal{C}^{1+\alpha}$, $r-1 \geq \alpha > 0$, splitting $T_xM = E^s(x) \oplus E^u(x)$ and constants $C, \lambda > 1$ such that for $n \geq 0$*

$$
\begin{aligned}
\|DF^n v\| &< C\lambda^{-n}\|v\| \qquad \text{if } v \in E^s;\\
\|DF^n v\| &< C\lambda^{n}\|v\| \qquad \text{if } v \in E^u.
\end{aligned}
\tag{2.3}
$$

As already mentioned we assume that the stable distribution E^s is orientable and that F preserves such an orientation. Also, to further simplify matters, we assume that F is transitive.

Next, we consider a flow ϕ_t generated by a vector field V satisfying the following properties.

Definition 2.3. *Let the vector field V be such that*

(i) $V \in \mathcal{C}^{1+\alpha}(\mathbb{T}^2, \mathbb{R}^2)$;
(ii) $\|V\| \in \mathcal{C}^r(\mathbb{T}^2, \mathbb{R}_{\geq})$;
(iii) for all $x \in \mathbb{T}^2$, $V(x) \neq 0$;
(iv) for each $x \in \mathbb{T}^2$, *we have* $V(x) \in E^s(x)$.

Remark 2.4. *Note that assumption (ii) essentially implies that we are just considering $\mathcal{C}^r$ time reparametrizations of the case $\|V\| = 1$. Hence, we are treating all the $\mathcal{C}^r$ reparametrizations on the same footing. This is rather interesting due to the scarcity of results on reparametrization of parabolic flows (see [16] for recent advances). In particular, it would be interesting to explore the applicability of the present strategy to the study of the mixing speed of the flow.*

Remembering (1.6), (1.3) and using the definition (1.7),

$$
\begin{aligned}
H_{x,t}(g) &= \int_0^t ds\, g \circ F^{-n} \circ \phi_{\tau_n(x,s)}(F^n(x))\\
&= \int_0^{\tau_n(x,t)} ds_1\, \nu_n(F^{-n} \circ \phi_{s_1}(F^n(x)))^{-1} g \circ F^{-n} \circ \phi_{s_1}(F^n(x)).
\end{aligned}
\tag{2.4}
$$

It is then natural to introduce the *transfer operator* [9] $\mathcal{L}_F \in L(\mathcal{C}^0, \mathcal{C}^0)$,

$$
\begin{aligned}
\mathcal{L}_F(g) :=& (\nu_1 \circ F^{-1})^{-1} g \circ F^{-1} = g \circ F^{-1} \frac{\|V\|}{\langle \widehat{V}, (DF\widehat{V}) \circ F^{-1}\rangle \, \|V\| \circ F^{-1}}\\
=& g \circ F^{-1} \frac{\|V\|}{\|DFV\| \circ F^{-1}} = g \circ F^{-1} \frac{\|DF^{-1}V\|}{\|V\| \circ F^{-1}},
\end{aligned}
\tag{2.5}
$$

where $\widehat{V}(x) = \|V(x)\|^{-1}V(x)$. We can now write

$$
H_{x,t}(g) = \int_0^{\tau_n(x,t)} ds_1\, (\mathcal{L}_F^n g)(\phi_{s_1}(F^n(x))) = H_{F^n(x), \tau_n(x,t)}(\mathcal{L}_F^n g).
\tag{2.6}
$$

The above formula is quite suggestive: if we fix $n = n_t(x)$ such that $\tau_{n_t}(x,t)$ is of order one, then, for each $x \in \mathbb{T}^2$ and $t \in \mathbb{R}_{\geq}$, $H_{x,t}(g)$ is expressed in terms of very similar functionals of $\mathcal{L}_F^{n_t} g$. In particular, such functionals are uniformly bounded on $\mathcal{C}^0$. It is thus natural to expect that, to address the questions put forward in

[9] Given a map F, in general a transfer operator associated to F has the form $\varphi \to \varphi \circ F^{-1} e^{\phi}$ for some function ϕ. Normally, the factor e^{ϕ} is called the *weight* while ϕ is the *potential*, [2].

the introduction, it suffices to understand the behaviour of $\mathcal{L}_F^n$ for large n. This obviously is determined by the spectral properties of $\mathcal{L}_F$.

Unfortunately, it is well known that the spectrum of $\mathcal{L}_F$ depends strongly on the Banach space on which it acts. For example, in the case of Remark 1.7 where F is a toral automorphism ϕ_t a rigid translation with unit speed, $\lambda^{-1}\mathcal{L}_F$ acting on L^2 is an isometry, hence the spectrum of $\mathcal{L}_F$ consists of the circle of radius $e^{h_{\text{top}}} = \lambda$; while, if we consider $\mathcal{L}_F$ acting on $\mathcal{C}^r$, the spectral radius will be given by λ^{r+1}.

This seems to render completely hopeless the above line of thought.

Yet, as mentioned in the introduction, it is possible to define anisotropic Banach spaces $\mathcal{C}^{p+q} \subset \mathcal{B}^{p,q} \subset (\mathcal{C}^q)'$, $p \in \mathbb{N}^*, q \in \mathbb{R}_{\geq}$, $p + q \leq r$, such that each transfer operators with $\mathcal{C}^r$ weight can be continuously extended to $\mathcal{B}^{p,q} = \overline{\mathcal{C}^\infty}^{\|\cdot\|_{p,q}}$, for some appropriate norm $\|\cdot\|_{p,q}$. The above are spaces of distributions, which can be annoying, but have several remarkable properties: i) the transfer operator, defined on $\mathcal{C}^r$, extends by continuity to a bounded operator on $\mathcal{B}^{p,q}$; ii) the transfer operator is a quasi-compact operator with a simple maximal eigenvalue; iii) the essential spectral radius of the transfer operator decreases exponentially with $\inf\{q,p\}$.[10]

The possibility to make the essential spectrum arbitrarily small, by increasing p and q, will play a fundamental role in our subsequent analysis. Unfortunately, a further problem now arises: the weight of $\mathcal{L}_F$ contains the vector field V which, by hypothesis, is only $\mathcal{C}^{1+\alpha}$. Hence $\mathcal{L}_F$ leaves $\mathcal{C}^r$ invariant only for $r \leq 1+\alpha$ (exactly the range in which we are not interested). Again it seems that we cannot use our strategy in any profitable manner.

Yet, such a problem has been overcame as well, e.g., in [22]. The basic idea is to extend the dynamics F to the oriented Grassmannian. Indeed, looking at (2.5), it is clear that the weight can be essentially interpreted as the expansion of a volume form. The simplest idea would then be to let the dynamics act on one forms on $\mathbb{T}^2$. Unfortunately, the length cannot be written exactly as a volume form on $\mathbb{T}^2$,[11] hence the convenience of being a bit more sophisticated: the weight of $\mathcal{L}_F$ can be written as the expansion of a one dimensional volume form on the space containing V. As V is exactly the tangent vector to the curves along which we integrate, we are led to consider functions on the Grassmannian made by one dimensional subspaces. However, in the simple case at hand, the construction in [22] can be considerably simplified. Namely, we can limit ourselves to considering the compact set $\Omega_* = \{(x,v) \in \mathbb{T}^2 \times \mathbb{R}^2 \;:\; \|v\| = 1,\, v \in \overline{C^s(x)}\}$. Moreover, since we have assumed that the stable distribution is orientable, then Ω_* is the disjoint union of two sets (corresponding to the two possible orientations). Let Ω be the connected component that contains the elements $(x, \widehat{V}(x))$. In addition, since we have also assumed that F preserves the orientation of the stable distribution, calling $\mathbb{F}$ the projectivization of F_* we have $\Omega_0 = \mathbb{F}^{-1}(\Omega) \subset \Omega$.

[10] Before [8] it was unclear if spaces with such properties existed at all. Nowadays there exists a profusion of possibilities. We use the ones stemming from [21, 22] because they seem particularly well suited for the task at hand, but any other possibility (e.g. [4, 5, 7]) should do.

[11] Yet, one can obtain something uniformly proportional to the length, hence one could use the spaces detailed in [20], which would be well suited also for studying the geodesic versus horocyclic flows case. We prefer the following choice because it is a bit cleaner and it can be used for more general operators.

Thus we have that $\mathbb{F} : \Omega_0 \subset \Omega \to \Omega$ is defined as

$$\mathbb{F}(x, v) = (F(x), \|D_x F v\|^{-1} D_x F v),$$
$$\mathbb{F}^{-1}(x, v) = (F^{-1}(x), \|D_x F^{-1} v\|^{-1} D_x F^{-1} v).$$

Also note that

$$\mathbb{F}^{-1}(x, \widehat{V}(x)) = (F^{-1}(x), \widehat{V}(F^{-1}(x))). \tag{2.7}$$

Thus, if we define the natural extension $\boldsymbol{\phi}_t(x, v) = (\phi_t(x), \|D_x \phi_t v\|^{-1} D_x \phi_t v)$, we have the analogous of (1.6)

$$\mathbb{F}^n(\boldsymbol{\phi}_s(x, \widehat{V}(x))) = \boldsymbol{\phi}_{\tau_n(x,s)}(\mathbb{F}^n(x, \widehat{V}(x))). \tag{2.8}$$

We can now define the transfer operator associated to $\mathbb{F} : \mathcal{C}^0(\Omega_0, \mathbb{R}) \to \mathcal{C}^0(\Omega, \mathbb{R})$ as

$$\mathcal{L}_{\mathbb{F}} \boldsymbol{g}(x, v) = \boldsymbol{g} \circ \mathbb{F}^{-1}(x, v) \frac{\langle \widehat{\boldsymbol{\pi}} \mathbb{F}^{-1}(x, v), D_x F^{-1} v \rangle \, \|V(x)\|}{\|V \circ F^{-1}(x)\|}, \tag{2.9}$$

where we have introduced the projections $\boldsymbol{\pi}(x, v) = x$, $\widehat{\boldsymbol{\pi}}(x, v) = v$.

Remark 2.5. *Note that $\mathbb{F}$ is itself a hyperbolic map with the repellor $\{(x, v) \in \Omega \ : \ v = \widehat{V}(x)\}$ and it has an invariant splitting of the tangent space with two dimensional unstable distribution and one dimensional stable.*

We can then define, for each $\boldsymbol{g} \in \mathcal{C}^0(\Omega, \mathbb{R})$, $t \in \mathbb{R}_{\geq}$ and $x \in \mathbb{T}^2$, the new functional

$$\mathbb{H}_{x,t}(\boldsymbol{g}) = \int_0^t ds \, \boldsymbol{g}(\phi_s(x), \widehat{V}(\phi_s(x)) \tag{2.10}$$

and easily obtain the analogous of (2.6) for the operator $\mathcal{L}_{\mathbb{F}}$.

Lemma 2.6. *For each $\boldsymbol{g} \in \mathcal{C}^0(\Omega, \mathbb{R})$, $n \in \mathbb{N}$, $t \in \mathbb{R}_{\geq}$ and $x \in \mathbb{T}^2$ we have*

$$\mathbb{H}_{x,t}(\boldsymbol{g}) = \mathbb{H}_{F^n x, \tau_n(x,t)}(\mathcal{L}_{\mathbb{F}}^n \boldsymbol{g}).$$

Proof. The proof is by direct computation:

$$\begin{aligned}
\mathbb{H}_{x,t}(\boldsymbol{g}) &= \int_0^t ds \, \boldsymbol{g}(\phi_s(x), \widehat{V}(\phi_s(x)) \\
&= \int_0^{\tau_n(x,t)} ds_1 \, \nu_n(\mathbb{F}^{-n} \circ \boldsymbol{\phi}_{s_1}(\mathbb{F}^n(x, \widehat{V}(x))))^{-1} \boldsymbol{g} \circ \mathbb{F}^{-n} \circ \boldsymbol{\phi}_{s_1} \circ \mathbb{F}^n(x, \widehat{V}(x)) \\
&= \int_0^{\tau_n(x,t)} ds_1 (\mathcal{L}_{\mathbb{F}}^n \boldsymbol{g})(\boldsymbol{\phi}_{s_1} \circ \mathbb{F}^n(x, \widehat{V}(x))) = \mathbb{H}_{F^n x, \tau_n(x,t)}(\mathcal{L}_{\mathbb{F}}^n \boldsymbol{g}).
\end{aligned}$$

□

The key observation is then that, for each function $g \in \mathcal{C}^r(\mathbb{T}^2, \mathbb{R})$, if we define $\boldsymbol{g} = \boldsymbol{\pi}^* g := g \circ \boldsymbol{\pi}$, then $\boldsymbol{g} \in \mathcal{C}^r(\Omega, \mathbb{R})$ and

$$\mathcal{L}_{\mathbb{F}} \boldsymbol{g}(x, \widehat{V}(x)) = \mathcal{L}_F g(x).$$

The above shows that understanding the properties of $\mathcal{L}_{\mathbb{F}}$ yields a control on $\mathcal{L}_F$. In addition, from definition (2.9) it is apparent that $\mathcal{L}_{\mathbb{F}}(\mathcal{C}^{r-1}(\Omega, \mathbb{R})) \subset \mathcal{C}^{r-1}(\Omega, \mathbb{R})$. We have thus completely eliminated the above mentioned regularity problem.

The basic fact about the operator $\mathcal{L}_{\mathbb{F}}$ is that there exists Banach spaces $\mathcal{B}^{p,q}$,[12] detailed in Appendix A to which $\mathcal{L}_{\mathbb{F}}$ can be continuously extended.[13] Moreover in Appendix A we prove the following result.

Proposition 2.7. *Let $F \in \mathcal{C}^r(\mathbb{T}^2, \mathbb{T}^2)$ be an Anosov map. Let $p \in \mathbb{N}^*$ and $q \in \mathbb{R}$ such that $p + q \leq r - 1$ and $q > 0$. Let $\rho = \exp(h_{top})$ where h_{top} is the topological entropy of F. Then the spectral radius of $\mathcal{L}_{\mathbb{F}}$ on $\mathcal{B}^{p,q}$ is ρ and its essential spectral radius is at most $\rho\lambda^{-\min\{p,q\}}$. In addition, ρ is a simple eigenvalue of $\mathcal{L}_{\mathbb{F}}$ and all the other eigenvalues are strictly smaller in norm.*[14]

We are finally ready to state precisely our first result.

Theorem 2.8. *Provided r is large enough,*[15] *the obstructions $\{O_i\}_{i=1}^{N_1} \subset \mathcal{C}^r(\mathbb{T}^2, \mathbb{R})'$ exist and are exactly the natural restrictions of a base of the eigenspaces associated to the discrete eigenvalues $\{\rho_i\}_{i\geq 1}$ of $\mathcal{L}_{\mathbb{F}}$ when acting of $\mathcal{B}^{p,q}$ for appropriate p, q, $p + q \leq r - 1$. Let d_i be the dimension of the eigenspace associated to ρ_i and set $D_i = \sum_{j\leq i} d_j$. Thus, there exists $C > 0$ such that, if $g \in \mathcal{C}^r$, $O_i(g) = 0$ for all $i < k \leq N_1$ and $O_k(g) \neq 0$, then, for all $x \in \mathbb{T}^2$,*

$$\left| \int_0^t g \circ \phi_s(x) ds \right| \leq C t^{\alpha_k} (\ln t)^{b_k}$$

and, for some $\bar{x} \in \mathbb{T}^2$,

$$\left| \int_0^t g \circ \phi_s(\bar{x}) ds \right| \geq C^{-1} t^{\alpha_k} (\ln t)^{b_k}$$

where, for $i > 0$ and $k \in (D_{i-1}, D_i]$, we have $\alpha_k = \frac{\ln |\rho_i|}{h_{top}}$ for and $b_k \in [0, d_i]$. Also $\alpha_{N_1} = b_{N_1} = 0$.

The next step is to study, in the case $O_i(g) = 0$ for all $i \in \{1, \dots, N_1\}$, the regularity of the coboundary. As already mentioned, see Remark 1.5, it is natural to consider only $r_i \leq 1 + \alpha$. Also to study exactly the Hölder regularity is a bit cumbersome so, in the spirit of giving ideas rather than detailed results, we content ourselves with the following.

Theorem 2.9. *Provided r is large enough,*[16] *there exist distributions $\{O_i\}_{i=N_1+1}^{N_2} \subset \mathcal{C}^r(\mathbb{T}^2, \mathbb{R})'$ such that if $O_i(g) = 0$ for all $i \in \{1, \dots, N_2\}$, then g is a Lipschitz coboundary. In addition, O_{N_1+1} is the SRB measure of F^{-1} and there exists Banach spaces $\widehat{\mathcal{B}}^{p,q}$ and a transfer operator $\widehat{\mathcal{L}}_{\mathbb{F}}$, determined by the map F, such that the $\{O_i\}_{i=N_1+2}^{N_2}$ are exactly the natural restrictions of a base of the eigenspaces associated to the discrete eigenvalues $\{\rho_i\}$ of $\widehat{\mathcal{L}}_{\mathbb{F}}$ when acting of $\widehat{\mathcal{B}}^{p,q}$ for appropriate p, q, $p + q \leq r - 2$.*

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[12] These are more general with respect to the previously mentioned ones. We give them the same name to simplify notation and since no confusion can arise.

[13] By a slight abuse of notations we will still call $\mathcal{L}_{\mathbb{F}}$ such an extension.

[14] The conditions on p, q are not optimal. The lack of optimality begin due to the fact that we require $p \in \mathbb{N}$. See [5], and reference therein, for different approaches that remove such a constraint.

[15] For example, $e^{h_{\text{top}}} \lambda^{-r/2} < 1$ suffices. We refrain to give a more precise characterisation of the minimal r since, in the present context, it is not very relevant.

[16] Here r needs to be much larger than in the previous Theorem. An estimate is implicit in the proof, but the reader may be better off thinking that F is $\mathcal{C}^\infty$ and not worrying about this issue.

The Banach spaces $\widehat{\mathcal{B}}^{p,q}$ are a bit more complex than the $\mathcal{B}^{p,q}$ used in Theorem 2.8 insofar they are really spaces of currents rather then distributions (one has to think of dg, rather than g, as an element of the Banach space). A part from this, the logic of the proofs of our two theorems is exactly the same.

The rest of the paper is devoted to the proof of the above claims.

3. Growth of the ergodic average

There is one further, and luckily last, obstacle to implementing the above mentioned strategy: the functionals $\mathbb{H}_{x,t}$ are, in general, not continuous (let alone uniformly continuos) on $\mathcal{B}^{p,q}$.[17] It is indeed possible to introduce Banach spaces on which the transfer operator is quasi-compact and the functionals $\mathbb{H}_{x,t}$ are continuos (this are spaces developed to handle piecewise smooth dynamics such as [9, 6, 10, 3]) but the essential spectral radius would be rather large and hence we would not be able to obtain all the relevant distributions. It turns out to be more convenient to introduce, for each $x \in \mathbb{T}^2$ and $\varphi \in L^\infty(\mathbb{R}_\geq, \mathbb{R})$, the new "mollified" functional

$$\mathbb{H}_{x,\varphi}(\boldsymbol{g}) = \int_{\mathbb{R}} \varphi(t) \cdot \boldsymbol{g} \circ \boldsymbol{\phi}_t(x, \widehat{V}(x))\, dt. \tag{3.1}$$

It is proven in Appendix A that $\mathbb{H}_{x,\varphi} \in (\mathcal{B}^{p,q})'$ provided $\varphi \in \mathcal{C}_0^{p+q}(\mathbb{R},\mathbb{R})$. The basic idea of our approach is that the above functionals suffice for our purposes. To be more precise let us define the sets $\mathcal{D}^s_{r,t,C} = \{\varphi \in \mathcal{C}^r([0,t],\mathbb{R}) \; : \; \|\varphi\|_{\mathcal{C}^r} \leq C\}$ and $\mathcal{D}_{r,C} = \{\varphi \in \mathcal{C}^r_0(\mathbb{R}_\geq,\mathbb{R}) \; : \; \|\varphi\|_{\mathcal{C}^r} \leq C\}$, note that such sets are locally compact in $\mathcal{C}^{r-1}([0,t],\mathbb{R})$ and $\mathcal{C}_0^{r-1}(\mathbb{R}_\geq,\mathbb{R})$, respectively.[18]

Lemma 3.1. *There exists $C_* > 0$ such that, for each $n \in \mathbb{N}$, $t \in \mathbb{R}_\geq$, $x \in \mathbb{T}^2$ and $\boldsymbol{g} \in \mathcal{C}^{r-1}(\Omega,\mathbb{R})$, there exists $K \in \mathbb{N}$, $\{n_i^\pm\}_{i=1}^K \subset \mathbb{N}$, $n_K^\pm = 0$, and $\{\varphi_i^\pm\}_{i=1}^K \subset \mathcal{D}_{r,C_*}$, $\{\varphi^\pm\} \subset \mathcal{D}^s_{r,t,C_*}$ such that*

$$\mathbb{H}_{x,t}(\boldsymbol{g}) = \sum_{\sigma\in\{+,-\}} \sum_{i=1}^K \mathbb{H}_{F^{n_i^\sigma}(x),\varphi_i^\sigma}(\mathcal{L}_{\mathbb{F}}^{n_i^\sigma}\boldsymbol{g}) + \mathbb{H}_{x,\varphi^\sigma}(\boldsymbol{g}).$$

Moreover, $\max\{|\operatorname{supp}\varphi^\pm|, |\operatorname{supp}\varphi_i^\pm|\} \leq 1$.

Proof. Fix $x \in \mathbb{T}^2$ and $t \in \mathbb{R}_\geq$. Let $n_t = \inf\{n \in \mathbb{N} \; : \; \tau_n(x,t) < 1\}$. By definition $\tau_{n_t}(x,t) \in (\Lambda^{-1}, 1)$ for some fixed $\Lambda > 1$. Note that, by [20, Lemma C.3], we have

$$\frac{\ln t}{h_{\text{top}}} - C_\# \leq n_t \leq \frac{\ln t}{h_{\text{top}}} + C_\#. \tag{3.2}$$

Let $\delta \in (0, \Lambda^{-1}/4)$ small and $C_* > 0$ large enough to be fixed later. We can now fix $n_1 = n_t$ and chose $\psi \in \mathcal{D}_{r,C_*/2}$ such that $\operatorname{supp}\psi \subset (\delta, \tau_{n_1} - \delta)$, $\psi|_{[2\delta,\tau_{n_1}-2\delta]} = 1$. Set $\psi^- = (1-\psi)\mathbb{1}_{[0,\tau_{n_1}/2]}$, $\psi^+ = (1-\psi)\mathbb{1}_{[\tau_{n_1}/2,\tau_{n_1}]}$. Then $\psi^\pm \in \mathcal{D}^s_{r,\tau_{n_1},C_*}$ and we can use Lemma 2.6 to write

$$\begin{aligned}\mathbb{H}_{x,t}(\boldsymbol{g}) &= \mathbb{H}_{F^{n_1}(x),\tau_{n_1}(x,t)}(\boldsymbol{g})\\ &= \mathbb{H}_{F^{n_1}(x),\psi^-}(\mathcal{L}_{\mathbb{F}}^{n_1}\boldsymbol{g}) + \mathbb{H}_{F^{n_1}(x),\psi}(\mathcal{L}_{\mathbb{F}}^{n_1}\boldsymbol{g}) + \mathbb{H}_{F^{n_1}(x),\psi^+}(\mathcal{L}_{\mathbb{F}}^{n_1}\boldsymbol{g}).\end{aligned}$$

[17] The problem comes from the boundary in the domain of the integral defining them. There, in some sense, the integrand jumps to zero and cannot be considered smooth in any effective manner.

[18] Up to now the exact definition of the $\mathcal{C}^r$ norms was irrelevant, now instead it does matter. We make the choice $\|\varphi\|_{\mathcal{C}^r} = \sum_{k=0}^r 2^{r-k}\|\varphi^{(k)}\|_{L^\infty}$. It is well known that with such a norm $\mathcal{C}^r$ is a Banach algebra.

We are happy with the middle term which, by Lemma A.4, is a continuous functional of $\mathcal{L}_{\mathbb{F}}^{n_1} g$, not so the other two terms. We have thus to take care of them. A computation analogous to the one done in Lemma 2.6 yields, for each $n \in \mathbb{N}$,

$$\mathbb{H}_{x,\varphi\circ\tau_n(x,\cdot)}(\boldsymbol{g}) = \mathbb{H}_{F^n(x),\varphi}(\mathcal{L}_{\mathbb{F}}^n \boldsymbol{g}). \tag{3.3}$$

We will use the above to prove inductively the formula

$$\begin{aligned} \mathbb{H}_{x,t}(\boldsymbol{g}) =& \mathbb{H}_{F^{n_k^-}(x),\psi_k^-}(\mathcal{L}_{\mathbb{F}}^{n_k^-}\boldsymbol{g}) + \mathbb{H}_{F^{n_k^+}(x),\psi_k^+}(\mathcal{L}_{\mathbb{F}}^{n_k^+}\boldsymbol{g}) \\ &+ \sum_{\sigma\in\{+,-\}}\sum_{i=1}^{k}\mathbb{H}_{F^{n_i^\sigma}(x),\varphi_i^\sigma}(\mathcal{L}_{\mathbb{F}}^{n_i^\sigma}\boldsymbol{g}) \end{aligned} \tag{3.4}$$

where $\psi_1^\pm = \psi^\pm$, $\varphi_1^\pm = \frac{1}{2}\psi$, $n_1^\pm = n_1$, $\psi_k^\pm \in \mathcal{D}^s_{r,\tau_{n_k^\pm},C_*}$, $\{\varphi_i^\pm\} \subset \mathcal{D}_{r,C_*}$, $\|\psi_k^\pm\|_{L^\infty} \le 1$, $\|\varphi_k^\pm\|_{L^\infty} \le 1$, $(b_k^\pm, b_k^\pm \mp \delta) \subset \operatorname{supp}\psi_k^\pm \subset (b_k^\pm, b_k^\pm \mp 2\delta)$, $\operatorname{supp}\varphi_k^\pm \subset (b_k^\pm, b_k^\pm \mp 1)$, $b_k^- = 0$ and $b_k^+ \in [0, \Lambda^{n_k^+}]$, $b_1 = t$.

Let us consider the first term on the right hand side of the first line of (3.4) (the second one can be treated in total analogy). Let $\operatorname{supp}(\psi_k^-) = [0, a_k]$ and define $m+1 = \inf\{n \in \mathbb{N} \;:\; \tau_n^{-1}(F^{n_k^-}(x), a_k) \ge 1\}$. Note that, by construction, there exists a fixed $\overline{m} \in \mathbb{N}$ such that $m \ge \overline{m}$, also $\overline{m}$ can be made large by choosing δ small. Hence $\widehat{\psi}_k^-(s) = \psi_k^- \circ \tau_m(F^{n_k^-}(x), s)$ is supported in the interval $[0,1)$ and the support contains $[0, \Lambda^{-1}]$.

Next, we need an estimate on the norm of $\widehat{\psi}_k^-$. We state it in a sub-lemma so the reader can easily choose to skip the, direct but rather tedious, proof.

Sub-Lemma 3.2. *Provided we choose δ small and C_* large enough, we have*

$$\widehat{\psi}_k^- \in \mathcal{D}_{r,\tau_{n_{k+1}^-},C_*/2},$$

where $n_{k+1}^- = n_k^- - m$.

Proof. First of all $\|\psi_k^-\|_{L^\infty} \le 1$, and[19]

$$\begin{aligned} \|\widehat{\psi}_k^-\|_{\mathcal{C}^r} &\le \sum_{j=0}^{r} 2^{r-j}\|\psi_k^-\|_{\mathcal{C}^j}\|\tilde{\nu}_{z_k,m}\|_{\mathcal{C}^{r-1}}\|\tilde{\nu}_{z_k,m}\|_{\mathcal{C}^{r-2}}\cdots\|\tilde{\nu}_{z_k,m}\|_{\mathcal{C}^{r-j}} \\ &\le 2^r + C_*\sum_{j=1}^{r}\|\tilde{\nu}_{z_k,m}\|_{\mathcal{C}^{r-1}}\|\tilde{\nu}_{z_k,m}\|_{\mathcal{C}^{r-2}}\cdots\|\tilde{\nu}_{z_k,m}\|_{\mathcal{C}^{r-j}} \end{aligned} \tag{3.5}$$

where $z_k = F^{n_k^-}(x)$ and, for each $j \in \mathbb{N}$ and $z \in \mathbb{T}^2$, $\tilde{\nu}_{z,m}(s) = \nu_m(\phi_s(z))$. To continue, we compute

$$\begin{aligned} \frac{d}{ds}\tilde{\nu}_{z_k,m}(s) &= \tilde{\nu}_{z_k,m}(s)\sum_{l=0}^{m-1}\frac{\langle\nabla\nu_1(F^l\circ\phi_s(z_k)), V(F^l\circ\phi_s(z_k))\rangle}{\nu_1(F^l\circ\phi_s(z_k))}\tilde{\nu}_{z_k,l}(s) \\ &= \tilde{\nu}_{z_k,m}(s)\sum_{l=0}^{m-1}\left[\frac{\langle V,\nabla\nu_1\rangle}{\nu_1}\right]\circ\phi_{\tau_l(z_k,s)}(F^l(z_k))\tilde{\nu}_{z_k,l}(s). \end{aligned}$$

[19] Here we use the formula $\|f\circ g\|_{\mathcal{C}^r} \le \sum_{k=0}^{r} 2^{r-k}\|f\|_{\mathcal{C}^k}\|Dg\|_{\mathcal{C}^{r-1}}\|Dg\|_{\mathcal{C}^{r-2}}\cdots\|Dg\|_{\mathcal{C}^{r-k}}$, that can be verified by induction.

The above, by induction, implies that there exist increasing constants $A_q \geq 1$ such that $\|\tilde\nu_{z_k,m}\|_{\mathcal{C}^q} \leq A_q \|\tilde\nu_{z_k,m}\|_{\mathcal{C}^0}$. Indeed, $[\frac{\langle V, \nabla\nu_1\rangle}{\nu_1}] \circ \phi_\cdot \in \mathcal{C}^{r-1}$, and

$$\begin{aligned}\left\|\left[\frac{\langle V,\nabla\nu_1\rangle}{\nu_1}\right]\circ\phi_{\tau_l(z_k,\cdot)}(F^l(z_k))\right\|_{\mathcal{C}^q} &\leq \sum_{i=0}^{q} 2^{q-i}\left\|\left[\frac{\langle V,\nabla\nu_1\rangle}{\nu_1}\right]\circ\phi_\cdot(F^l(z))\right\|_{\mathcal{C}^q}\|\tilde\nu_{F^l(z_k),l}\|^i_{\mathcal{C}^{q-1}}\\ &\leq C_\# \sum_{i=0}^{q} A^i_{q-1}\lambda^{-il} \leq C_\# A^q_{q-1}.\end{aligned}$$

Thus,

$$\begin{aligned}\|\tilde\nu_{z_k,m}\|_{\mathcal{C}^{q+1}} &= 2^q\|\tilde\nu_{z_k,m}\|_{\mathcal{C}^0} + \|\frac{d}{ds}\tilde\nu_{z_k,m}\|_{\mathcal{C}^q}\\ &\leq 2^q\|\tilde\nu_{z_k,m}\|_{\mathcal{C}^0} + \|\tilde\nu_{z_k,m}\|_{\mathcal{C}^q}\sum_{l=0}^{m-1} C_\# A^q_{q-1}A_{q-1}\lambda^{-l}\\ &\leq \left[2^q + A_q C_\# A^{q+1}_{q-1}\right]\|\tilde\nu_{z_k,m}\|_{\mathcal{C}^0} =: A_{q+1}\|\tilde\nu_{z_k,m}\|_{\mathcal{C}^0}.\end{aligned}$$

We did not try to optimise the above computation since the only relevant point is that the A_q do not depend on m. Accordingly, if we choose δ small (and hence m large) enough, we have $\|\tilde\nu_{z_k,m}\|_{\mathcal{C}^q} \leq \frac14$ for all $q \leq r-1$. Using the above fact in (3.5) yields

$$\|\widehat\psi^-_k\|_{\mathcal{C}^r} \leq 2^r + C_*\sum_{j=1}^{r} 4^{-j} = 2^r + \frac13 C_*$$

which implies the Lemma provided we choose C_* large. □

By (3.3), we have

$$\mathbb{H}_{F^{n^-_k}(x),\psi^-_k}(\mathcal{L}^{n^-_k}_{\mathbb{F}}\boldsymbol{g}) = \mathbb{H}_{F^{n^-_{k+1}}(x),\widehat\psi^-_k}(\mathcal{L}^{n^-_{k+1}}_{\mathbb{F}}\boldsymbol{g}).$$

Again we can write $\psi^-_{k+1} = (1-\psi)\widehat\psi^-_k\mathbb{1}_{[0,2\delta]}$ and $\varphi^-_{k+1} = \widehat\psi^-_k - \psi^-_{k+1}$. Then,

$$\sup\{\|\psi^-_{k+1}\|_{\mathcal{C}^r([0,1],\mathbb{R})}, \|\varphi^-_{k+1}\|_{\mathcal{C}^r_0(\mathbb{R},\mathbb{R})}\} \leq C_*$$

and $[0,\delta] \subset \operatorname{supp}\psi^-_{k+1} \subset [0,2\delta]$. Accordingly

$$\mathbb{H}_{F^{n^-_k}(x),\psi^-_k}(\mathcal{L}^{n^-_k}_{\mathbb{F}}\boldsymbol{g}) = \mathbb{H}_{F^{n^-_{k+1}}(x),\psi^-_{k+1}}(\mathcal{L}^{n^-_{k+1}}_{\mathbb{F}}\boldsymbol{g}) + \mathbb{H}_{F^{n^-_{k+1}}(x),\varphi^-_{k+1}}(\mathcal{L}^{n^-_{k+1}}_{\mathbb{F}}\boldsymbol{g}).$$

The Lemma is thus proven by taking $k = K$, so that $n^\pm_K = 0$.[20] □

We are now ready to prove our first result.

Proof of Theorem 2.8. By Proposition 2.7 we have

$$\mathcal{L}^n_{\mathbb{F}} = \sum_{j=0}^{m} \rho^n_j Q^n_j + R^n_{p,q} \tag{3.6}$$

where m is a finite number, ρ_j, $|\rho_{j+1}| \leq |\rho_j| \leq e^{h_{\rm top}}$, are complex eigenvalues of $\mathcal{L}_{\mathbb{F}}$, Q_j are finite rank operators with spectrum equal to $\{0,1\}$ and $R_{p,q}$ is a linear operator with spectral radius at most $e^{\beta_{\rm ess}} = \lambda^{-\min(p,q)}e^{h_{\rm top}}$. Moreover,

[20] If more steps are needed on one side, say the plus side, one can simply set $\psi^-_{k+1} = \psi^-_k$ and $\varphi^-_k = 0$ for all the extra steps.

$Q_j \cdot R_{p,q} = R_{p,q} \cdot Q_j = 0$ and $Q_i Q_j = Q_j Q_i = 0$. Let d_j be the dimension of the range of Q_j and $J_j + 1$ be the size of the maximal Jordan block contained in Q_j. Finally, let $\{\Pi_{i,j}\}_{j=1}^{d_j}$ be commuting rank one projections such that $\mathcal{L}_{\mathbb{F}}$, restricted to the range of $\sum_{j\le k} \Pi_{i,j}$ has a maximal Jordan Block of size $b_k + 1 \le J_j + 1$ and $\sum_{j\le d_i} \Pi_{i,j}$ is the spectral projection on the range of Q_i. Finally, set[21]

$$\alpha_j = \frac{\ln |\rho_j|}{h_{\rm top}}\ ; \quad N_1 = \sum_{j=1}^m d_j.$$

We can then choose p, q and m such that $\beta_{\rm ess} < 0$ and $|\rho_j| \ge 1$ for all $j \le m$. Then, equation (3.2) and Lemmata 3.1 and A.4 imply that,

$$\begin{aligned} H_{x,t}(g) &= \mathcal{O}(\|g\|_{L^\infty}) + \sum_{n=1}^{n_t}\left[\sum_{j=0}^m \mathcal{O}(\rho_j^n \|Q_j^n \boldsymbol{g}\|_{p,q}) + \mathcal{O}(\|R_{p,q}^n \boldsymbol{g}\|_{p,q})\right] \\ &= \mathcal{O}(\|g\|_{L^\infty}) + \sum_{j=0}^m\sum_{k=1}^{d_j}\sum_{n=1}^{n_t} \mathcal{O}(\rho_j^n n^{b_k} \|\Pi_{j,k}\boldsymbol{g}\|_{p,q}) + \sum_{n=1}^{n_t}\mathcal{O}(\|g\|_{\mathcal{C}^r} e^{n\beta_{\rm ess}/2}) \qquad (3.7)\\ &= \sum_{j=0}^m\sum_{k=1}^{d_j} (\ln t)^{b_k} \mathcal{O}(t^{\alpha_j}\|\Pi_{j,k}\boldsymbol{g}\|_{p,q}) + \mathcal{O}(\|g\|_{\mathcal{C}^r}). \end{aligned}$$

To conclude note that, since the $\Pi_{j,k}$ are rank one, they must be of the form $h_i \otimes O_i$ with $h_i \in \mathcal{B}^{p,q}$ and $O_i \in (\mathcal{B}^{p,q})' \subset (\mathcal{C}^r)'$. In addition, by Lemma A.4 the h_i cannot be in the kernel of all the relevant functional. The above immediately implies the wanted result. □

4. Coboundary regularity

In the previous section we have seen that if g belongs to the kernel of enough O_i, then the $H_{x,t}(g)$ are all uniformly bounded. We promised in the introduction to prove that the same is then true for $\overline{H}_{x,t}(g)$, now is the time to do it.

Lemma 4.1. *If g belongs to the kernel of the distributions $\{O_i\}_{i=1}^{N_1}$, then the functions $\overline{H}_{x,t}(g)$ are uniformly bounded. If, in addition, g has zero average with respect to the SRB measure of F^{-1}, then*

$$\lim_{T\to\infty}\int_0^T dt\, \chi' \circ \tau_{n_T}(x,t)\nu_{n_T}(\phi_t(x)) g\circ\phi_t(x) = 0$$

uniformly in x.

Proof. The idea is to repeat the same exact arguments done in the previous section in the present, slightly more general, context. We refrained from doing the argument directly in the needed generality only for didactical purposes. Equation (3.3) implies that

$$\mathbb{H}_{x,\chi\circ\tau_n(x,\cdot)}(\boldsymbol{g}) = \mathbb{H}_{F^n(x),\chi}(\mathcal{L}_{\mathbb{F}}^n \boldsymbol{g}).$$

We can then obtain a representation of $\mathbb{H}_{x,\chi\circ\tau_n(x,\cdot)}(\boldsymbol{g})$ of the form stated in Lemma 3.1 since the proof holds verbatim for the present case. It immediately follows that the $\mathbb{H}_{x,\chi\circ\tau_n(x,\cdot)}(\boldsymbol{g})$, and hence the $\overline{H}_{x,t}(g)$, are uniformly bounded arguing like in the proof of Theorem 2.8.

[21] Note that $\alpha_1 = 1$, Q_1 is a one dimensional projection and $\alpha_i < 1$ for all $i > 0$.

As for the second statement, similarly to (2.4), we have

$$\int_0^T dt\,\chi'\circ\tau_{n_T}(x,t)\nu_{n_T}(\phi_t(x))g\circ\phi_t(x) = \int_0^{\tau_{n_T}(x,t)} \chi'(s)g\circ F^{-n_T}(\phi_s(F^n(x))) = \int_0^1 \chi'(s)g\circ F^{-n_T}(\phi_s(F^n(x))).$$

Note that, by definition $\chi' \in \mathcal{C}_0^{r-1}([0,1],\mathbb{R})$ and it is thus a nice test function. We are thus in a much simpler situation than the one discussed in the previous section: we have directly a good functional and the relevant transfer operator, let us call it $\mathcal{L}_{0,F}$, it is simply the composition with F^{-1}. Hence $\mathcal{L}_{0,F}$ acts nicely on $\mathcal{C}^r$ with no further complications. This situation could be handled by the simpler Banach spaces introduced in [21], since the dual of $\mathcal{L}_{0,F}$ is the standard Ruelle-Perron-Frobenius operator for F^{-1}. Yet, we can apply the arguments of the Appendix A as well. It turns out that $\mathcal{L}_{0,F}$ has one as the maximal eigenvalue and a spectral gap. Moreover, the left eigenvector associated to the eigenvalue one is the SRB measure of F^{-1}. Hence if $\mu_{\mathrm{SRB}}(g) = 0$, then the integral vanishes as an inverse power of t. □

As discussed in the introduction, Lemma 4.1 implies that g is a measurable coboundary. To study its regularity we must investigate the regularity of $\overline{H}_{x,t}(g)$ with respect to x. For simplicity we will discuss only the first derivative, since, after all, this is just a simple example and it cannot be expected to reflect exactly the behavior of more interesting cases which, typically, carry some extra structure. First of all recall that given a one form $\omega(x) = \sum_{i=1}^2 a_i(x)dx_i$ and diffeomorphism $G \in \mathcal{C}^1(\mathbb{T}^2,\mathbb{T}^2)$ the pullback form is given by

$$G^*\omega(x) = \sum_i a_i(G(x))(D_xG)_{ij}dx_j,$$

while for a vector field v the pushforwad is given by

$$G_*v(x) = D_{G^{-1}(x)}G\cdot v(G^{-1}(x)).$$

Using such a notation, by a direct computation we have that, for each $v\in\mathbb{R}^2$,

$$v_i\partial_{x_i}H_{x,t}(g) = \int_0^t ds(\partial_{x_k}g)\circ\phi_s(x)(D_x\phi_s)_{k,i}v_i = \int_0^t ds\,(dg)_{\phi_s(x)}((\phi_s)_*v) \tag{4.1}$$

where we have used the usual convention on the summation of repeated indexes. As done before, we would like to use the renormalizing dynamics F to transform the above expression in integrals over uniformly bounded domains. Unfortunately, now the algebra is a bit more involved. To start with we must differentiate (1.6):

$$\begin{aligned} D_{\phi_t(x)}F^n\cdot D_x\phi_t &= D_{F^n(x)}\phi_{\tau_n(x,t)}\cdot D_xF^n + V(\phi_{\tau_n(x,t)}(F^n(x)))\otimes\nabla\tau_n(x,t)\\ &= D_{F^n(x)}\phi_{\tau_n(x,t)}\cdot D_xF^n\left[\mathbb{1}+\nu_n(x)^{-1}V(x)\otimes\nabla\tau_n(x,t)\right].\end{aligned}$$

Hence, setting $A_{x,t,n} = \mathbb{1}+\nu_n(x)^{-1}V(x)\otimes\nabla\tau_n(x,\tau_n^{-1}(x,t))$, we have

$$D_x\phi_t = D_{\phi_{\tau_n(x,t)}\circ F^n(x)}F^{-n}\cdot D_{F^n(x)}\phi_{\tau_n(x,t)}\cdot D_xF^n\cdot A_{x,\tau_n(x,t),n}. \tag{4.2}$$

Inserting (4.2) in (4.1) and changing variables we obtain, for each $v\in\mathbb{R}^2$,

$$\langle v,\nabla H_{x,t}(g)\rangle = \int_0^{\tau_n(x,t)} ds\,\frac{\left[(F^{-n})^*dg\right]_{\phi_s\circ F^n(x)}((\phi_s)_*F^n_*A_{x,s,n}v)}{\nu_n(F^{-n}\circ\phi_s(F^n(x)))}. \tag{4.3}$$

The equation is now assuming a form similar to (2.4) and it seems then possible to express it again in terms of an appropriate transfer operator, this time acting on one forms. Yet, in order to do so effectively we need more informations on the relevant objects.

Our first step consists in finally substantiating the assertion that ϕ_t is a *parabolic* dynamics.

Lemma 4.2. *There exists $C, \beta > 0$ such that, for all $x \in \mathbb{T}^2$ and $t \in \mathbb{R}_\geq$, letting $\xi(s) = D_x\phi_s$, we have*

$$\|\xi\|_{\mathcal{C}^{r-1}((0,t),GL(2,\mathbb{R}))} \leq Ct^\beta.$$

Proof. It turns out to be convenient to define $V^\perp(x)$ as the perpendicular vector to $V(x)$ such that $\|V^\perp(x)\| = \|V(x)\|^{-1}$. In this way we can use $\{V(x), V^\perp(x)\}$ as basis of the tangent space at x, and the changes of variable are uniformly bounded, with determinant one and $\mathcal{C}^r$ in the flow direction. In such coordinates we have

$$D_x\phi_t = \begin{pmatrix} 1 & a(x,t) \\ 0 & b(x,t) \end{pmatrix}.$$

Confronting with formula (4.2) yields, for each $n \in \mathbb{N}$,

$$b(x,t) = \langle V^\perp(\phi_t(x)), D_{\phi_{\tau_n(x,t)}\circ F^n(x)}F^{-n} \cdot D_{F^n(x)}\phi_{\tau_n(x,t)} \cdot D_xF^nV^\perp(x)\rangle.$$

We then choose n so that $\tau_n \in [\Lambda^{-1}, 1]$, hence n is proportional to $\ln t$. By compactness it follows that $\|D_{F^n(x)}\phi_{\tau_n(x,t)}\| \leq C_\#$. Hence there exists $\beta_0 > 0$ such that

$$\sup_{x\in\mathbb{T}^2} |b(x,t)| \leq C_\# t^{\beta_0}.$$

On the other hand, by the semigroup property, for each $m \in \mathbb{N}$,

$$D_x\phi_m = \prod_{i=0}^{m-1} \begin{pmatrix} 1 & a(\phi_i(x),1) \\ 0 & b(\phi_i(x),1) \end{pmatrix} = \begin{pmatrix} 1 & \sum_{j=0}^{m-1} a(\phi_j(x),1)b(x,m-j) \\ 0 & b(x,m) \end{pmatrix}.$$

Since, again by compactness, $|a(x,1)| \leq C_\#$, it follows

$$|a(x,m)| \leq C_\# \sum_{j=0}^{m-1} |b(x,m-j)| \leq C_\# \sum_{j=0}^{m-1} j^{\beta_0} \leq C_\# m^{\beta_0+1}.$$

Hence

$$\|\xi\|_{\mathcal{C}^0((0,t),GL(2,\mathbb{R}))} \leq C_\# t^\beta$$

with $\beta = \beta_0 + 1$.

To estimate the derivative notice that $\dot{\xi}(s) = D_{\phi_s(x)}V\xi(s)$. To understand the regularity of the above equation, recall that the stable foliation can be expressed in local coordinates by $(x_1, G(x_1,x_2))$, where $G \in \mathcal{C}^r$, $G(0,x_2) = x_2$, so that $\{(x_1, G(x_1,x_2))\}_{x_1\in\mathbb{R}}$ is the leaf through the point $x = (x_1,x_2)$, and $(1, \partial_{x_1}G(x)) = V(x)$. It is known that, in such coordinates, $\partial_{x_2}G(\cdot,x_2) \in \mathcal{C}^{r-1}$ uniformly, see [23] and references therein. Then, by Schwarz Lemma, if follows that $\partial_{x_2}\partial_{x_1}G(\cdot,x_2) \in \mathcal{C}^{r-2}$. Hence, $DV \circ \phi_t$ is a $\mathcal{C}^{r-2}$ function of t, with uniformly bounded norm. Accordingly, for each $k \in \{0,\dots,r-2\}$,

$$\begin{aligned}\|\xi\|_{\mathcal{C}^{k+1}((0,t),GL(2,\mathbb{R}))} &\leq \|\dot{\xi}\|_{\mathcal{C}^k((0,t),GL(2,\mathbb{R}))} + 2^{k+1}\|\xi\|_{\mathcal{C}^0((0,t),GL(2,\mathbb{R}))} \\ &\leq C_\#\|\xi\|_{\mathcal{C}^k((0,t),GL(2,\mathbb{R}))},\end{aligned}$$

from which the Lemma readily follows. □

Next, we need to spell out the cocyle properties of τ_n and

$$\Theta_{x,n}(s) = D_xF^n \cdot A_{x,s,n}. \tag{4.4}$$

Lemma 4.3. *For each $x \in \mathbb{T}^2$, $n, m \in \mathbb{N}$ and $s \in \mathbb{R}_{\geq}$ we have that $\Theta_{x,m}(s)$ is invertible and*

$$\begin{aligned}&\tau_m(F^n(x), \tau_n(x,s)) = \tau_{n+m}(x,s)\\&\Theta_{F^n(x),m}(\tau_m(F^n(x),s)) \cdot \Theta_{x,n}(s) = \Theta_{x,n+m}(\tau_m(F^n(x),s)).\end{aligned}$$

Proof. Since DF and $D\phi_t$ are invertible by definition, the first assertion is equivalent to the fact that $A_{x,\tau_n(x,t),n}$ is invertible. If not, then there would exists $v \in \mathbb{R}^2$ such that

$$\left[\mathbb{1} + \nu_n(x)^{-1} V(x) \otimes \nabla\tau_n(x,t)\right] v = 0.$$

The above implies that $v = cV(x)$ for some $c \in \mathbb{R}$. Hence

$$-\nu_n(x) = \langle \nabla\tau_n(x,t), V(x)\rangle = \int_0^t \frac{d}{ds}\nu_n(\phi_s(x))ds = \nu_n(\phi_t(x)) - \nu_n(x),$$

which is impossible since $\nu_n \neq 0$. The proof of the other equalities consists of a boring computation: by definition

$$\begin{aligned}\tau_m(F^n(x), \tau_n(x,t)) &= \int_0^{\tau_n(x,t)} \nu_m(\phi_s(F^n(x)))ds = \int_0^t \nu_m(\phi_{\tau_n(x,s)}(F^n(x)))\nu_n(\phi_s(x))ds\\&= \int_0^t \nu_m(F^n \circ \phi_s(x))\nu_n(\phi_s(x))ds = \int_0^t \nu_{n+m}(\phi_s(x))ds = \tau_{n+m}(x,t),\end{aligned}$$

where we have used (1.6). Next, using again the definition,

$$\begin{aligned}\Theta_{F^n(x),m}(\tau_m(F^n(x),s)) \cdot \Theta_{x,n}(s) = &D_{F^n(x)}F^m \left[\mathbb{1} + \frac{1}{\nu_m(F^n(x))} V(F^n(x)) \otimes \nabla\tau_m(F^n(x),s)\right]\\&\times D_xF^n \left[\mathbb{1} + \frac{1}{\nu_n(x)} V(x) \otimes \nabla\tau_n(x, \tau_n^{-1}(x,s))\right]\\=&D_xF^{n+m} \left[\mathbb{1} + \frac{1}{\nu_{n+m}(x)} V(x) \otimes (D_xF^n)^*\nabla\tau_m(F^n(x),s)\right]\\&\times \left[\mathbb{1} + \frac{1}{\nu_n(x)} V(x) \otimes \nabla\tau_n(x, \tau_n^{-1}(x,s))\right].\end{aligned}$$

To continue we remark that, differentiating the first equality of the Lemma, yields

$$\nabla\tau_{n+m}(x,s) = (D_xF^n)^*\nabla\tau_m(F^n(x), \tau_n(x,s)) + \nu_m(\phi_{\tau_n(x,s)}(F^n(x)))\nabla\tau_n(x,s).$$

Thus,

$$\begin{aligned}\Theta_{F^n(x),m}(\tau_m(F^n(x),s))\Theta_{x,n}(s) = &D_xF^{n+m}\Bigg[\mathbb{1} + \frac{1}{\nu_{n+m}(x)} V(x) \otimes \nabla\tau_{n+m}(x, \tau_n^{-1}(x,s))\\&- \frac{\nu_m(\phi_s \circ F^n(x))}{\nu_{n+m}(x)} V(x) \otimes \nabla\tau_n(x, \tau_n^{-1}(x,s)) + \frac{1}{\nu_n(x)} V(x) \otimes \nabla\tau_n(x, \tau_n^{-1}(x,s))\\&+ \frac{\langle \nabla\tau_{n+m}(x, \tau_n^{-1}(x,s)), V(x)\rangle}{\nu_{n+m}(x)\nu_n(x)} V(x) \otimes \nabla\tau_n(x, \tau_n^{-1}(x,s))\\&- \frac{\nu_m(\phi_s \circ F^n(x))\langle \nabla\tau_n(x, \tau_n^{-1}(x,s)), V(x)\rangle}{\nu_{n+m}(x)\nu_n(x)} V(x) \otimes \nabla\tau_n(x, \tau_n^{-1}(x,s))\Bigg].\end{aligned}$$

To conclude note that, for each $k \in \mathbb{N}$ and $r \in \mathbb{R}_{\geq}$,

$$\langle \nabla \tau_k(x,r), V(x)\rangle = \int_0^r \frac{d}{ds}\nu_k(\phi_s(x))ds = \nu_k(\phi_r(x)) - \nu_k(x).$$

Using such an equality in the above formula, and remembering that

$$\tau_{n+m}(x, \tau_n^{-1}(x,s)) = \tau_m(F^n(x), s),$$

the Lemma follows. □

Finally we need bounds on $\Theta_{x,n}$.

Lemma 4.4. *There exists $C, \beta_1 > 0, \Lambda_0 > 1$ such that, for each $n \in \mathbb{N}$ and $x \in \mathbb{T}^2$, we have*

$$\|\Theta_{x,n}\|_{\mathcal{C}^{r-2}((0,t),GL(2,\mathbb{R}))} \leq C\Lambda_0^n \tau_n^{-1}(x,t)^{\beta_1}.$$

Proof. By Lemma 4.2 and equation (4.2) it follows

$$\|\Theta_{x,n}\|_{\mathcal{C}^{r-2}((0,t),GL(2,\mathbb{R}))} \leq C_\# \|DF^n\|_{\mathcal{C}^r} \tau_n^{-1}(x,t)^\beta t^\beta,$$

which implies the Lemma. □

The above shows that (4.3) can be bounded in terms of norms of transfer operators acting on forms. Of course, we still have the same regularity problem that appeared in the previous sections, which will be dealt with in the same way, i.e. extending the dynamics to the Grassmanian. We can then define the operator, acting on one forms $\mathfrak{g}$ defined on Ω by

$$\left[\widehat{\mathcal{L}}_{\mathbb{F}}\mathfrak{g}\right]_{(x,v)} = \frac{\langle \widehat{\boldsymbol{\pi}}\mathbb{F}^{-1}(x,v), D_x F^{-1} v\rangle \, \|V(x)\|}{\|V \circ F^{-1}(x)\|} \left[(\mathbb{F}^{-1})^* \mathfrak{g}\right]_{(x,v)}. \tag{4.5}$$

Next we need to define again a more general functional. For each one form $\mathfrak{g}$ on Ω and vector field $w \in L^\infty(\mathbb{R}, \mathbb{R}^2)$, with compact support in $\mathbb{R}_\geq$, we define

$$\mathbb{H}^1_{x,w}(\mathfrak{g}) = \int_{\mathbb{R}} \mathfrak{g}_{\boldsymbol{\phi}_s(x,\hat V(x))}((D_x\phi_s w(s), 0))ds.$$

We can then write[22]

$$\begin{aligned}\langle v, \nabla H_{x,t}(g)\rangle &= \int_0^{\tau_n(x,t)} ds \left[\widehat{\mathcal{L}}^n_{\mathbb{F}} \boldsymbol{\pi}^* dg\right]_{\mathbb{F}^n(x,\hat V(x))} ((D_{F^n(x)}\phi_s \Theta_{x,n}(s)v, 0)) \\ &= \mathbb{H}^1_{F^n(x),\Theta_{x,n}v}(\widehat{\mathcal{L}}^n_{\mathbb{F}} \boldsymbol{\pi}^* dg).\end{aligned} \tag{4.6}$$

To conclude we need the analogous of equation (3.3) and Lemma 3.1. Let $\mathcal{A}$ be the set of one forms on Ω such that, for all $v \in \mathbb{R}^2$, $\mathfrak{g}((0,v)) = 0$.

Lemma 4.5. *For each $t \in \mathbb{R}_\geq$, one form $\mathfrak{g} \in \mathcal{A}$ and vector field $w \in L^\infty(\mathbb{R}, \mathbb{R}^2)$, with compact support, we have*

$$\mathbb{H}^1_{x,[\Theta_{x,n}^{-1}w]\circ\tau_n(x,\cdot)}(\mathfrak{g}) = \mathbb{H}^1_{F^n(x),w}(\widehat{\mathcal{L}}^n_{\mathbb{F}}\mathfrak{g}).$$

Proof. By hypothesis $\mathfrak{g}(v_1, v_2) = g_1(v_1)$ for some one form g_1 on $\mathbb{T}^2$. Then, by a computation similar to (4.3), we have

$$\begin{aligned}\mathbb{H}^1_{x,[\Theta_{x,n}^{-1}w]\circ\tau_n(x,\cdot)}(\mathfrak{g}) &= \int_{\mathbb{R}} [\phi_s^*(g_1)]_x([\Theta_{x,n}^{-1}w]\circ\tau_n(x,s))ds \\ &= \int_{\mathbb{R}} \nu_n(F^{-n}\circ\phi_s(F^n(x)))^{-1}[(F^{-n})^*g_1]_{\phi_s(F^n(x))}((\phi_s)_* w(s))ds.\end{aligned}$$

[22] Note that $(\mathbb{F}^{-n})^*\boldsymbol{\pi}^* dg = \boldsymbol{\pi}^*(F^{-n})^* dg$.

Next, note that if $\mathfrak{g} \in \mathcal{A}$, then also $(\mathbb{F}^n)^*\mathfrak{g} \in \mathcal{A}$, hence

$$\begin{aligned}\mathbb{H}^1_{x,[\Theta^{-1}_{x,n}w]\circ\tau_n(x,\cdot)}(\mathfrak{g}) &= \int_{\mathbb{R}} \frac{[(\mathbb{F}^{-n})^*\mathfrak{g})]_{\phi_s(F^n(x),\hat V(F^n(x)))}((\phi_s)_*w(s),0)}{\nu_n(F^{-n}\circ\phi_s(F^n(x)))}ds\\ &= \mathbb{H}^1_{F^n(x),w}(\widehat{\mathcal{L}}^n_{\mathbb{F}}\mathfrak{g}).\end{aligned}$$

□

We have finally obtained all the facts needed to conclude the argument in a way similar to what we did in Section 3. Let us define appropriate classes of vector fields: $\widehat{\mathcal{D}}^s_{r,C} = \{w : C^r(\mathbb{R}_\geq, \mathbb{R}^2) : \|w\|_{C^r} \leq C\}$ and $\widehat{\mathcal{D}}_{r,C} = \{w : C^r_0(\mathbb{R},\mathbb{R}^2) : \|w\|_{C^r} \leq C\}$. We are now ready to state the equivalent, in the present context, of the decomposition discussed in Lemma 3.1.

Lemma 4.6. *There exist $C_* > 0$ and $\Lambda_1 > 1$ such that, for each $n \in \mathbb{N}$, $t \in \mathbb{R}_\geq$, $v \in \mathbb{R}^2$, $\|v\| = 1$, and $g \in \mathcal{C}^r(\mathbb{T}^2,\mathbb{R})$, there exist $K \in \mathbb{N}$, $\{n_i\}_{i=1}^K \subset \mathbb{N}$, $n_K = 0$, and $w_i \subset \mathcal{D}_{r,C_*\Lambda_1^{n_i}}$, $w \in \mathcal{D}^s_{r,C_*}$ such that*

$$\langle v, \nabla H_{x,t}(g)\rangle = \sum_{i=1}^K \mathbb{H}^1_{F^{n_i^\sigma}(x),w_i}(\widehat{\mathcal{L}}^{n_i}_{\mathbb{F}}\mathfrak{g}) + \mathbb{H}^1_{x,w}(\mathfrak{g}).$$

Moreover, $\max\{|\operatorname{supp} w|, |\operatorname{supp} w_i|\} \leq 1$.

Proof. We use the exact same strategy of Lemma 3.1, only now we do not have to worry about the right extreme of the intervals since our test functions are automatically well behaved over there. We will discuss explicitly only the details which needs to be changed, since the proof given in Lemma 3.1 can be followed verbatim.

First of all note that $\mathfrak{g} = \boldsymbol{\pi}^* g \in \mathcal{A}$. Then, as in Lemma 3.1, but using Lemma 4.5, and using a ψ such that $\operatorname{supp}\psi \subset (\delta, 2)$, $\psi|_{[2\delta,1]} = 1$, $\psi^- = [1-\psi]\mathbb{1}_{[0,1]}$, we have

$$\langle v, \nabla \overline{H}_{x,t}(g)\rangle = \mathbb{H}^1_{F^{n_1}(x),\chi\psi^-\cdot\Theta_{x,n_1}v}(\widehat{\mathcal{L}}^{n_1}_{\mathbb{F}}\mathfrak{g}) + \mathbb{H}^1_{F^{n_1}(x),\chi\psi\cdot\Theta_{x,n_1}v}(\widehat{\mathcal{L}}^{n_1}_{\mathbb{F}}\mathfrak{g}).$$

Again, $\chi\psi\Theta_{x,n}v \in \mathcal{C}^{r-2}_0([0,1],\mathbb{R})$, hence it is a good test function. Not so for $\chi\psi^-\Theta_{x,n_1}v$ owing to the discontinuity at zero. To treat the latter term notice that, by Lemmata 4.3 and 4.5, setting $n_1 - n_2 = m$, we have

$$\begin{aligned}\mathbb{H}^1_{F^{n_1}(x),\chi\psi^-\cdot\Theta_{x,n_1}v}(\widehat{\mathcal{L}}^{n_1}_{\mathbb{F}}\mathfrak{g}) &= \mathbb{H}^1_{F^{n_2}(x),[\chi\psi^-\cdot\Theta^{-1}_{F^{n_2}(x),m}\Theta_{x,n_1}v]\circ\tau_m(F^{n_2}(x),\cdot)}(\widehat{\mathcal{L}}^{n_2}_{\mathbb{F}}\mathfrak{g})\\ &= \mathbb{H}^1_{F^{n_2}(x),[\chi\psi^-]\circ\tau_m(F^{n_2}(x),\cdot)\Theta_{x,n_2}v}(\widehat{\mathcal{L}}^{n_2}_{\mathbb{F}}\mathfrak{g}).\end{aligned}$$

The above formula shows that the construction can be carried out exactly as before. The needed estimates on the functions w_i, w follows from Lemma 4.4, computations similar to the ones in Sub-Lemma 3.2 and noticing that $\|\tau_m^{-1}\|_{\mathcal{C}^r} \leq C_\# \Lambda_2^m$ for some $\Lambda_2 > 1$. □

Having shown that the problem can be cast in a setting completely analogous to the one already discussed, we are now ready to conclude.

Proof of Theorem 2.9. This is the same argument carried out in the proof of Theorem 2.8, only now we need the spectral picture for the operator $\widehat{\mathcal{L}}_{\mathbb{F}}$ on an appropriate space $\widehat{\mathcal{B}}^{p,q}$, $p+q \leq r-2$. In appendix B we show that there exists a

Banach $\widehat{\mathcal{B}}^{p,q}$ on which $\widehat{\mathcal{L}}_{\mathbb{F}}$ has spectral radius $\rho > 0$ and essential spectral radius bounded by $\lambda^{-\min\{p,q\}}\rho$. Also the functionals $\mathbb{H}^1_{x,w}$, for $w \in \mathcal{C}^r_0$, are bounded by

$$|\mathbb{H}^1_{x,w}(\mathfrak{g})| \leq C_\# \|w\|_{\mathcal{C}^{r-2}} \|\mathfrak{g}\|_{p,q}.$$

Thus, using the spectral decomposition of $\widehat{\mathcal{L}}_{\mathbb{F}}$ it follows that, if r is large enough, we can choose p, q so that $\lambda^{-\min\{p,q\}}\rho =: e^{\beta_{\rm ess}} < \Lambda_1^{-1}$. It follows that if dg belongs to the kernel of all the distributions corresponding to the point spectrum of $\widehat{\mathcal{L}}_{\mathbb{F}}$, then Lemma 4.6 implies that

$$|\nabla H_{x,t}(g)| \leq C_\# \sum_{i=1}^{K} e^{\beta_{\rm ess} n_i} \Lambda_1^{n_i} \|g\|_{\mathcal{C}^r} + C_\# \|g\|_{\mathcal{C}^1} \leq C_\# \|g\|_{\mathcal{C}^r}.$$

This implies that the $\overline{H}_{x,t}(g)$ are equicontinuous functions of x, hence there exists $\{t_j\}$ such that $\overline{H}_{x,t_j}(g)$ converges uniformly to a Lipschitz function. We have thus showed that $\overline{H}_{x,t}(g)$ has a convergent subsequence to a Lipschitz function h which satisfies (1.12), hence g is a Lipschitz coboundary. □

Appendix A. Anisotropic Banach spaces:distributions

In this section we first construct the needed Banach spaces, then we discuss the relation with the Banach spaces constructed in [22], finally, we prove Proposition 2.7 and show that $\mathbb{H}$ is a bounded functional.

The construction of the Banach spaces are based on the definition of appropriate norms. The Banach spaces are then obtained by closing $\mathcal{C}^r(\Omega, \mathbb{C})$ with respect to such norms.[23] The basic idea is to control not the functions themselves but rather their integrals along curves close to the stable manifolds. Hence the first step is to define the set of relevant curves.[24] To do so we need to fix $\delta \in (0, 1/2)$ and $K \in \mathbb{R}_\geq$.

Definition A.1 (Admissible leaves)**.** *Given $r \in \mathbb{R}_\geq$, an admissible leave $W \subset \mathbb{T}^2$ is a $\mathcal{C}^r$ curve with length in the interval $[\delta/2, \delta]$. We require that there exists a parametrisation $\omega : [0, 1] \to W$ of such a curve such that $\omega'(\tau) \in C^s(\omega(\tau))$, for all $\tau \in [0, 1]$, and $\|\omega\|_{\mathcal{C}^r([0,1],\mathbb{T}^2)} \leq K$. Moreover we ask $(\omega(\tau), \omega'(\tau)\|\omega'(\tau)\|^{-1}) \in \Omega$, that is the curves have all the chosen orientation. We call Σ the set of admissible curves and, given $W \in \Sigma$, we will call ω_W any parametrisation satisfying the properties mentioned above.*

The above set is not empty and contains pieces of stable manifolds, if K is large enough, since the stable manifolds are uniformly $\mathcal{C}^r$, [24]. The basic fact about admissible curves is that if $W \in \Sigma$, then, for each $n \in \mathbb{N}$, $F^{-n}W \subset \cup_{i=1}^{N_n} W_i$ for some set $\{W_i\}_{i=1}^{N_n} \subset \Sigma$. This is quite intuitive but see [21] for a detailed proof in a more general setting.

Next, we define the integral of an element $\boldsymbol{g} \in \mathcal{C}^r(\Omega, \mathbb{C})$ along an element $W \in \Sigma$ agains any $\varphi \in \mathcal{C}^0(W, \mathbb{C})$:

$$\int_W \varphi \boldsymbol{g} := \int_0^1 ds\, \varphi \circ \omega_W(s) \cdot \boldsymbol{g}(\omega_W(s), \omega_W'(s)\|\omega_W'(s)\|^{-1})\|\omega_W'(s)\|. \tag{A.1}$$

[23] We consider complex valued functions because we are interested in having nice spectral theory.

[24] In fact, in the simple case at hand, we could consider directly pieces of stable manifolds. We do not do it to make easier to use already existing results.

Also, given $W \in \Sigma$ and $\varphi : W \to \mathbb{R}$ we set, for all $s \le r$,

$$\|\varphi\|_{\mathcal{C}^s(W,\mathbb{R})} \doteq \|\varphi \circ \omega_W\|_{\mathcal{C}^s([0,1],\mathbb{R})}. \tag{A.2}$$

We are now ready to define the relevant semi-norms:[25]

$$\|\boldsymbol{g}\|_{p,q} := \sup_{W\in\Sigma} \sup_{|\alpha|\le p} \sup_{\varphi\in\mathcal{C}_0^{q+|\alpha|}(W,\mathbb{C})} \int_W \varphi \cdot \partial^\alpha(\boldsymbol{g}), \tag{A.3}$$

where $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ is the usual multi–index and $1, 2$ refer to the x co-ordinate while 3 refers to θ. It is easy to check that the $\|\boldsymbol{g}\|_{p,q}$ are indeed semi-norms on $\mathcal{C}^r(\Omega, \mathbb{C})$.

Definition A.2 ($\mathcal{B}^{p,q}$ spaces)**.** *Let $p \in \mathbb{N}^*, q \in \mathbb{R}$, $p + q \le r$ and $q > 0$. We define $\mathcal{B}^{p,q}$ to be the closure of $\mathcal{C}^r(\Omega, \mathbb{C})$ with respect to the semi-norm $\|\cdot\|_{p,q}$.*[26]

Remark A.3. *Note that $\|\boldsymbol{g}\|_{p,q} \le \|\boldsymbol{g}\|_{\mathcal{C}^{p+q}}$.*

The Banach spaces defined above are well suited for the tasks at hand but, unfortunately, they are not exactly the one introduced in [22] where a more general theory is put forward. To avoid having to develop the theory from scratch, it is convenient to show how to relate the present setting to the one in [22]. To this end let us briefly recall the construction in [22], then we will explain the relation with the present one, so that we can apply the general results to the present context.

We start by recalling, particularizing them to our simple situation, the basic objects used in [22]: the r times differentiable sections $\mathcal{S}^r$ of a line bundle over the Grassmannian of one dimensional subspaces. More precisely, let $\mathcal{G} = \{(x, E)\}$ where $x \in \mathbb{T}^2$ and $E \subset \mathbb{R}^2$ is a linear one dimensional subspace, then $h \in \mathcal{S}^r$ is a $\mathcal{C}^r$ map $(x, E) \to E^*$.[27] Note that there is a strict relation between $\mathcal{S}^r$ and $\mathcal{C}^r(\Omega, \mathbb{C})$: for each $(x, v) \in \Omega$ let $E_v = \{\mu v\}_{\mu\in\mathbb{R}}$, then for each $h \in \mathcal{S}^r$ define $[\boldsymbol{i}h](x, v) = h(x, E_v)(v)$. The important fact is that the elements of $\mathcal{S}^r$ can be integrated along a curve in $\mathbb{T}^2$. Namely, let $W \in \Sigma$, $h \in \mathcal{S}^r$ and $\varphi \in \mathcal{C}^0(\mathbb{T}^2, \mathbb{C})$, then

$$\int_W \varphi h := \int_0^1 ds\, \varphi \circ \omega_W(s)\, h(\omega_W(s), E_{\omega'_W(s)})(\omega'_W(s)) = \int_W \varphi\, \boldsymbol{i}h.$$

Again the relevant norm in [22] is given by integrals along curves in Σ. Note that if $h, \tilde{h} \in \mathcal{S}^r$ differ only for (x, E) such that E does not belong to $C^s(x)$, then any norm of the difference based on integrations along curves in Σ will be zero. Thus, with respect to any such norm $\boldsymbol{i}$ will be an isomorphism. The readers can then check that the norms defined in [22] are equivalent to $\|\boldsymbol{i}h\|_{p,q}$. Thus our spaces $\mathcal{B}^{p,q}$ are isomorphic, as a Banach space, to the ones defined in [22] (that, not by chance, are again called $\mathcal{B}^{p,q}$). Finally, we have to understand how the operator $\mathcal{L}_{\mathbb{F}}$ reads in the corresponding language of [22]. To this end it is useful to introduce the operator $\Gamma : \mathcal{C}^r(\Omega, \mathbb{C}) \to \mathcal{C}^r(\Omega, \mathbb{C})$ defined by

$$(\Gamma\boldsymbol{g})(x, v) := \boldsymbol{g}(x, v)\|V(x)\|.$$

[25] By $\mathcal{C}^s_0(W, \mathbb{C})$ we mean the $\mathcal{C}^s$ functions with support contained in $\mathrm{Int}(W)$. The fact that the test functions must be zero at the boundary of W is essential for the following arguments.

[26] To be precise the elements of $\mathcal{B}^{p,q}$ are the equivalence classes determined by the equivalence relation $h \sim \bar{h}$ iff $\|h - \bar{h}\|_{p,q} = 0$.

[27] To be precise, since we are going to do spectral theory, we should consider the complex dual. We do not insist on this since the complexification is totally standard.

Note that, by the assumptions of Defintion 2.3, Γ is invertible and both the operator and its inverse can be extended to a continuous operator on $\mathcal{B}^{p,q}$. It then follows by equation (2.9) and [22, Section 3.2] that

$$\boldsymbol{i}^{-1}\Gamma^{-1}\mathcal{L}_{\mathbb{F}}\Gamma\boldsymbol{i}h = (F^{-1})^*h = F_*h, \tag{A.4}$$

that is $\mathcal{L}_{\mathbb{F}}$ is conjugated to the push-forward of F on $\mathcal{S}^r$.

Proof of Proposition 2.7. Since (A.4) states that our operator is conjugated to the push-forward F_*, all the spectral properties of F_* and $\mathcal{L}_{\mathbb{F}}$ coincide. It thus suffices to note that [22, Proposition 4.4, Theorem 5.1, Theorem 6.4] state that, for $q \in \mathbb{R}_>$, $p \in \mathbb{N}_>$ and $p+q \le r$, F_* can be extended continuously to $\mathcal{B}^{p,q}$, that the logarithm of the spectral radius of F_* is given by the topological entropy (which is the maxim of the metric entropy), that the maximal eigenvalue is simple and F_* has a spectral gap and the essential spectral radius is bounded by $e^{h_{\text{top}}}\lambda^{-\min\{p,q\}}$. □

We have thus seen that the operator $\mathcal{L}_{\mathbb{F}}$ acts very nicely on the spaces $\mathcal{B}^{p,q}$. The next important fact is that the functionals we are interested in are well behaved on such spaces.

Lemma A.4. *There exists $C > 0$ such that, for each $x \in \mathbb{T}^2$, $q \in \mathbb{R}_>$, $p \in \mathbb{N}_>$, $p+q \le r$, and $\varphi \in \mathcal{C}^r_0(\mathbb{R}_\ge, \mathbb{R})$, $\boldsymbol{g} \in \mathcal{C}^r(\Omega, \mathbb{R})$ we have*

$$|\mathbb{H}_{x,\varphi}(\boldsymbol{g})| \le C|\operatorname{supp}\varphi|\|\boldsymbol{g}\|_{p,q}\|\varphi\|_{\mathcal{C}^{p+q}}.$$

Moreover, if $\mathbb{H}_{x,\varphi}(\boldsymbol{g}) = 0$ for all x and φ, then $\boldsymbol{g} = 0$.

Proof. Let us start by considering the case $\operatorname{supp}\varphi \subset [a, a+\delta]$, for some $a > 0$. Then, $\{\phi_t(x)\}_{t\in[a,a+\delta]}$ is the re-parametrisation of a curve W in Σ. To see it just consider the parametrisation $\omega_W(s) = \phi_{a+\delta s}(x)$. Moreover, setting $\tilde\varphi(\phi_s(x)) = \varphi(s)$,[28] by (A.1) and (3.1)

$$\begin{aligned}\int_W \tilde\varphi\boldsymbol{g} &= \int_0^1 ds\ \varphi\circ\phi_{a+\delta s}(x)\boldsymbol{g}(\phi_{a+\delta s}(x), \widehat V(\phi_{a+\delta s}(x)))\|\widehat V(\phi_{a+\delta s}(x))\|\delta\\ &= \int_{\mathbb{R}} ds\ \varphi(s)(\Gamma\boldsymbol{g})\circ\boldsymbol{\phi}_s(x, \widehat V(x)) = \mathbb{H}_{x,\varphi}(\Gamma\boldsymbol{g}).\end{aligned}$$

Since the first quality on the left is exactly one of the functionals used in (A.3) to define the norm ($p=0$) and Γ^{-1} is a bounded operator on each space $\mathcal{B}^{p,q}$, we have

$$\|\mathbb{H}_{x,\varphi}(\boldsymbol{g})\| \le C_\#\|\Gamma^{-1}\|_{0,p}\|\varphi\|_{\mathcal{C}^q}\|\boldsymbol{g}\|_{0,q} \le C_\#\|\varphi\|_{\mathcal{C}^{p+q}}\|\boldsymbol{g}\|_{p,q}.$$

The Lemma follows then by using a partition of unity. The second statement follows by the above comment that the functionals $\mathbb{H}_{x,\varphi}$ are exactly the ones that are used to construct the $\|\cdot\|_{0,q}$ norm. Thus it must be $\|\boldsymbol{g}\|_{0,q} = 0$ and the result follows since the inclusion of $\mathcal{B}^{p,q}$ in $\mathcal{B}^{0,q}$ is one-one, [22]. □

Appendix B. Anisotropic Banach spaces: currents

In this appendix we briefly describe the Banach spaces of currents used in our second results and sketch the needed results. We will be much faster than in Appendix A and will omit several details as the construction is very similar an no essentially new ideas are present.

[28] Note that, since the stable manifolds are uniformly $\mathcal{C}^r$, [24], $\|\tilde\varphi\|_{\mathcal{C}^r(W,\mathbb{R})} \le C_\#\|\varphi\|_{\mathcal{C}^r(\mathbb{R}_\ge,\mathbb{R})}$.

We consider the same set of admissible leaves detailed in Definition A.1. For each $W \in \Sigma$, let $\mathcal{V}^q$ be the set of $\mathcal{C}^q$ vector fields compactly supported on W and with $\mathcal{C}^q$ norm bounded by one. Then, for each smooth one form $\mathfrak{g}$ defined on ω we define

$$\|\mathfrak{g}\|_{p,q} := \sup_{W\in\Sigma} \sup_{|\alpha|\leq p} \sup_{\varphi\in\mathcal{V}^{q+|\alpha|}} \int_W [\partial^\alpha(\mathfrak{g})](\varphi), \tag{B.1}$$

where the integral is defined as in the previous section.

Note that there exists a standard isomorphism $\boldsymbol{i}$ from vector fields to one forms, so that $\mathfrak{g}(\varphi) = \langle \mathfrak{g}, \boldsymbol{i}(\varphi)\rangle$. Thus the above norm is equivalent to the norm $\|\cdot\|_{p,q,1}$ used in [20]. Hence if we define $\widehat{\mathcal{B}}^{p,q}$ as the closure of the smooth one form with respect to the above norm, we obtain exactly the space $\mathcal{B}^{p,q,1}$ of [20].

Unfortunately, the transfer operator used here differs from the one studied in [20] insofar it has a potential, which was absent in [20]. In principle, we should therefore prove the Lasota-Yorke inequality for our operator and compute the spectral radius for the present operator via a variational principle (as in [22]). Since such a computation is completely standard but a bit lengthy, we just state a partial result for the reader convenience that suffices for our goals (in particular we do not compute exactly the spectral radius):

- The operator $\widehat{\mathcal{L}}_{\mathbb{F}}$ extends continuously on $\widehat{\mathcal{B}}^{p,q}$, has spectral radius ρ and essential spectral radius strictly bounded by $\lambda^{-\min\{p,q\}}\rho$.
- For all $w \in \mathcal{C}^r$ and $x \in \mathbb{T}^2$ and $p + q \leq r$ we have
$$\left|\mathbb{H}^1_{x,w}(\mathfrak{g})\right| \leq C_\#|\operatorname{supp} w|\|\mathfrak{g}\|_{p,q}\|w\|_{\mathcal{C}^{p+q}}.$$

The above two facts are all we presently needed.

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